

PRAKTIKUM 10

Binary Tree Traversal (In-order, Pre-order, Post-order)

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Kelas: (A/P2)

1. Kode Program

```
1 class Node:
2     def __init__(self, data):
3         self.left = None
4         self.right = None
5         self.data = data
6
7 def insert(root, data):
8     if root is None:
9         return Node(data)
10
11     if data < root.data:
12         root.left = insert(root.left, data)
13     else:
14         root.right = insert(root.right, data)
15     return root
16
17
18
19 def inorderTraversal(root):
20     if root:
21         inorderTraversal(root.left)
22         print(root.data, end=" ")
23
24 def preorderTraversal(root):
25     if root:
26         print(root.data, end=" ")
27         preorderTraversal(root.left)
28         preorderTraversal(root.right)
29
30 def postorderTraversal(root):
31     if root:
32         postorderTraversal(root.left)
33         postorderTraversal(root.right)
34         print(root.data, end=" ")
35
36
37 # =====
38 print("Nama : Syahrul Hidayatullah")
39 print("NIM : J0403251145")
40
41 root = Node(45)
42
43 data = [25,65,15,35,55,75,30]
44
45 for i in data:
46     insert(root,i)
47
48 print("\nIn-order Traversal:")
49 inorderTraversal(root)
50
51 print("\nPre-order Traversal:")
52 preorderTraversal(root)
53
54 print("\nPost-order Traversal:")
55 postorderTraversal(root)
```

2. Hasil Program

```
PS C:\Users\Syahrul Hidayatullah\Documents\kuliah\sem 2\algoritma> & "C:\Users\Syahrul Hidayatullah\AppData\Local\Programs\Python\Python313\python.exe" "c:/Users/Syahrul Hidayatullah/
ts/kuliah/sem 2/algoritma/pp10/pi0.py"
Nama : Syahrul Hidayatullah
NIM : J0403251145

In-order Traversal:
15 25 45
Pre-order Traversal:
45 25 15 35 30 65 55 75
Post-order Traversal:
15 30 35 25 55 75 65 45
PS C:\Users\Syahrul Hidayatullah\Documents\kuliah\sem 2\algoritma>
```

3. Hasil Traversal

In-order Traversal: 15, 25, 30, 35, 45, 55, 65, 75

Pre-order Traversal: 45, 25, 15, 35, 30, 65, 55, 75

Post-order Traversal: 15, 30, 35, 25, 55, 75, 65, 45

4. Penjelasan

a. Perbedaan Ketiga Traversal

In-order

Traversal kiri → root → kanan.

Pada BST menghasilkan data terurut dari kecil ke besar.

Pre-order

Traversal root → kiri → kanan.

Memperlihatkan struktur awal tree.

Post-order

Traversal kiri → kanan → root.

Umum digunakan saat menghapus node/tree.

b. Traversal yang paling mudah melihat struktur awal tree

Menurut saya Pre-order Traversal, karena root ditampilkan terlebih dahulu sehingga susunan awal pohon lebih mudah dipahami.

c. Apakah hasil traversal sama jika urutan data berbeda?

Tidak.

Jika urutan input berbeda maka bentuk BST dapat berubah, sehingga hasil traversal juga bisa berubah.